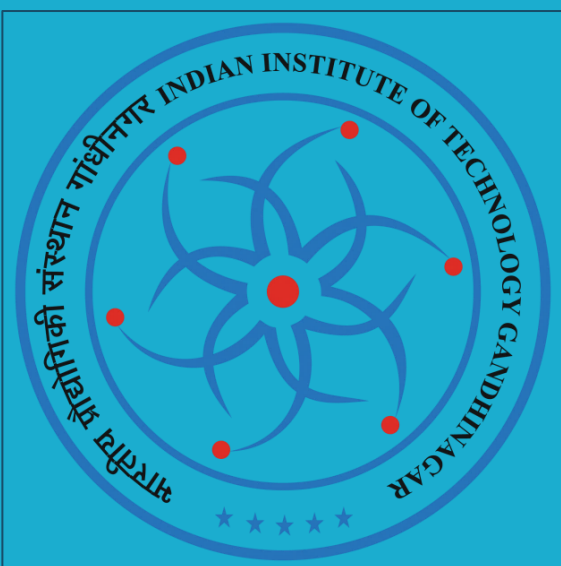




Impact of Structural Parameters on RF properties of GaAs HEMT

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Introduction

- HEMTs leverage a **heterojunction interface** to create a **high-mobility 2DEG** channel. By minimizing electron scattering, they enable ultra-fast switching and high gain, essential for **5G/6G** and **radar** technologies.
- Parameters being varied to optimize the **InGaAs HEMT** RF performance: **Gate Length**, **$\text{In}_x\text{Ga}_{1-x}$ Mole Fraction**, **Substrate Doping**, **Silicon Doping in Substrate**, **Buffer Layer**(Carbon Doping, Iron Doping).
- InGaAs Material Property:**

InGaAs Property	Value (Approx. for $x = 0.53$)
Electron Mobility	$> 10,000 \text{ cm}^2/V.s$
Saturation Velocity	$\approx 2.0 \times 10^7 \text{ cm/s}$
Effective Mass	$\approx 0.041 m_0$
Bandgap	0.35 to 1.42 eV
Thermal Conductivity	$\approx 0.05 \text{ W/cm.K}$

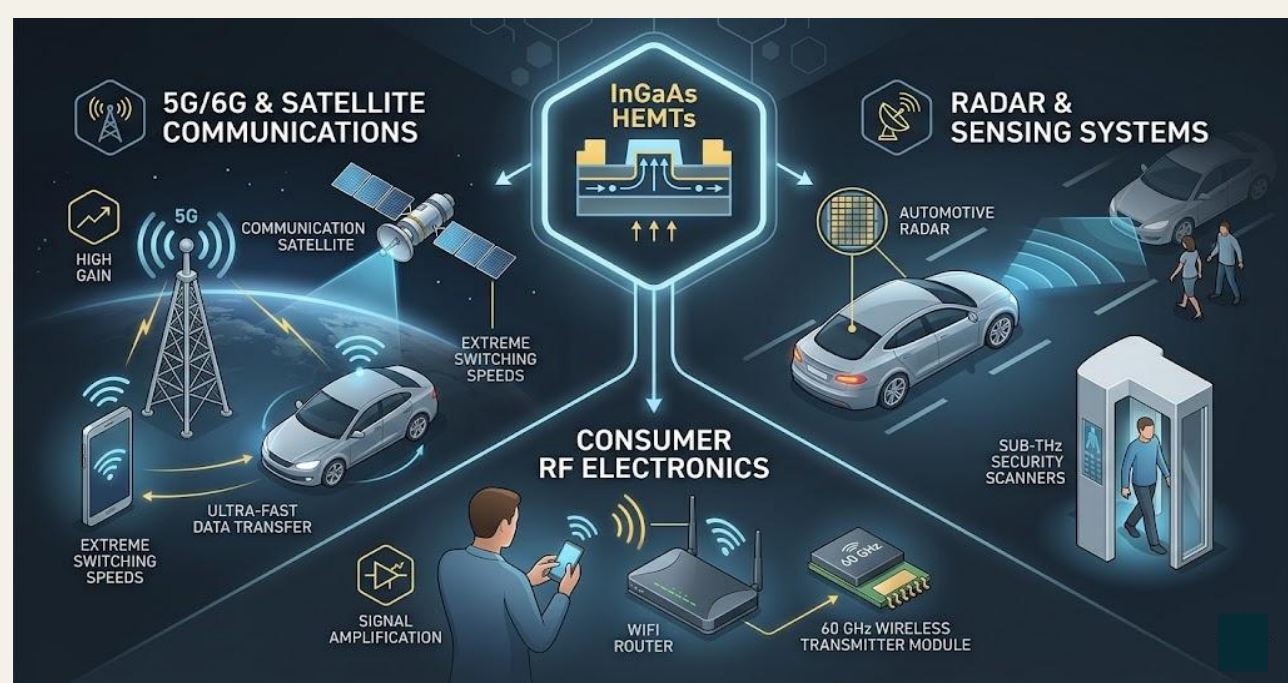


Figure 1: Applications of HEMT InGaAs

Device Structure

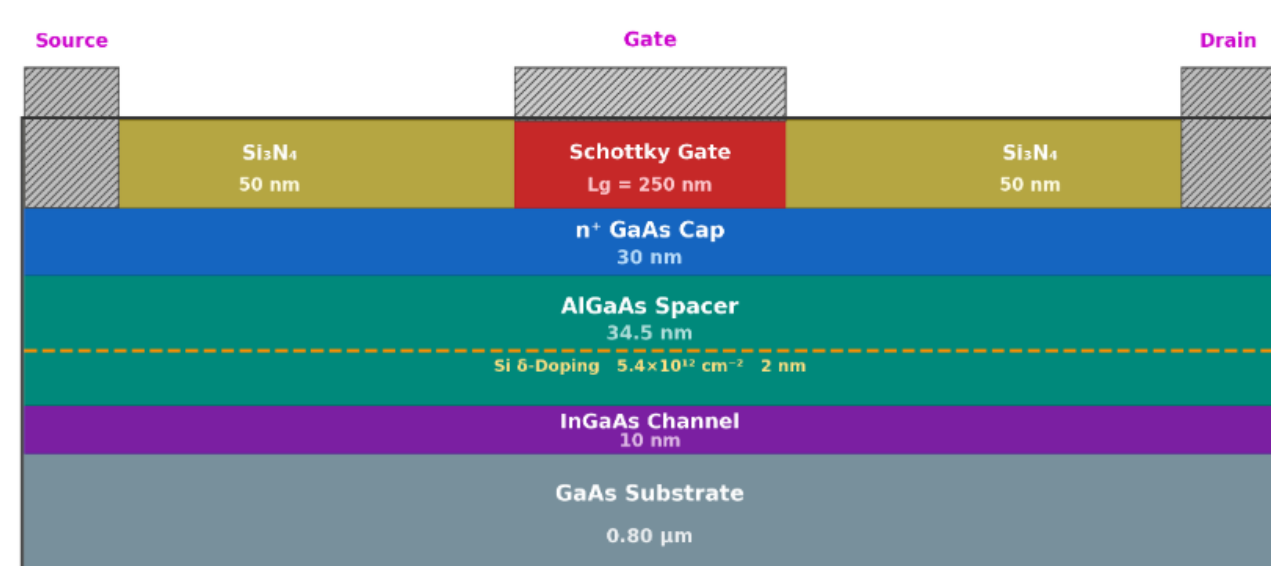


Figure 2: HEMT InGaAs Device Structure

Gate Length Variation

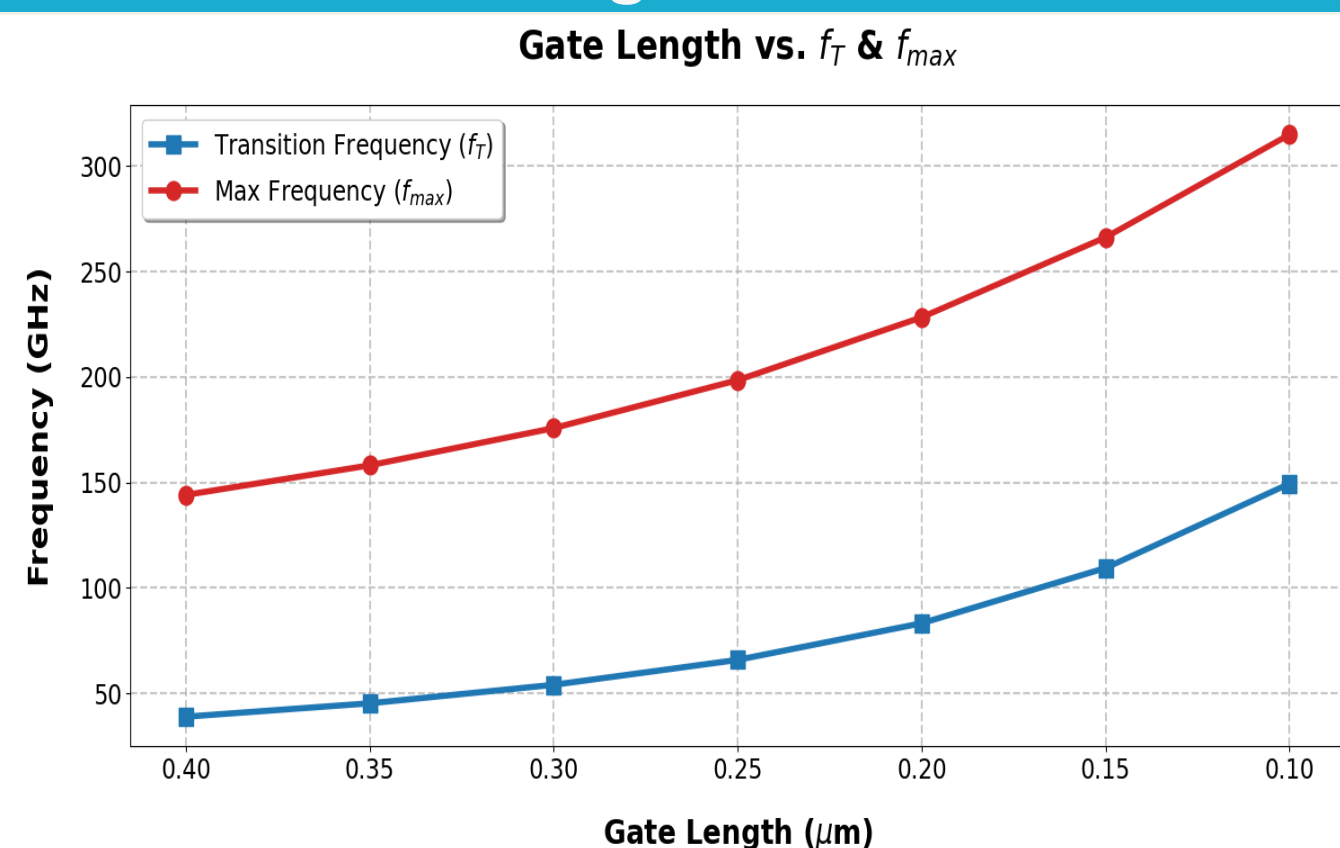


Figure 3: Gate Length vs Ft and Fmax

- Scaling Range:** Changing Gate Length from $0.40 \mu\text{m}$ down to $0.10 \mu\text{m}$.
- Effect:** Improves performance. It reduces electron transit time and increases transconductance (g_m).
- Observations:** Both f_T and f_{max} show an exponential-like increase as the gate is shortened, with f_{max} reaching its peak above 300 GHz .

$\text{In}_x\text{Ga}_{1-x}$ As Mole Fraction

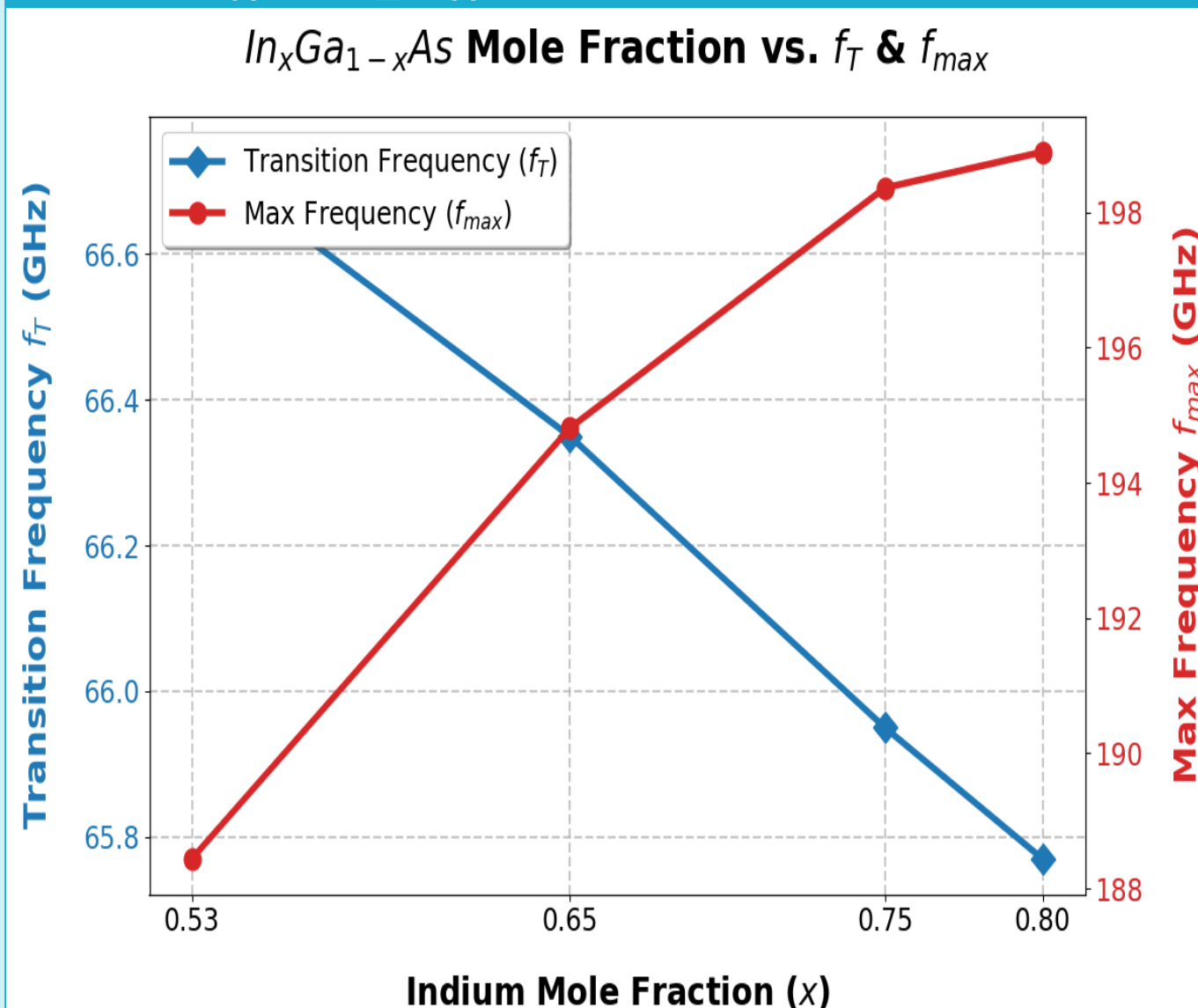


Figure 4: InGaAs Mole Fraction vs Ft and Fmax.

- Scaling Range:** Changing Mole Fraction(x) from 0.53 to 0.80 in the $\text{In}_x\text{Ga}_{1-x}$ As channel.
- Effect:** Improves electron mobility and saturation velocity, which boosts f_{max} . However, it slightly increases gate capacitance (C_g), causing a minor dip in f_T .
- Observations:** f_{max} rises from ≈ 188 to $\approx 199 \text{ GHz}$ due to higher mobility, while f_T dips slightly due to increased gate capacitance.

Substrate Doping

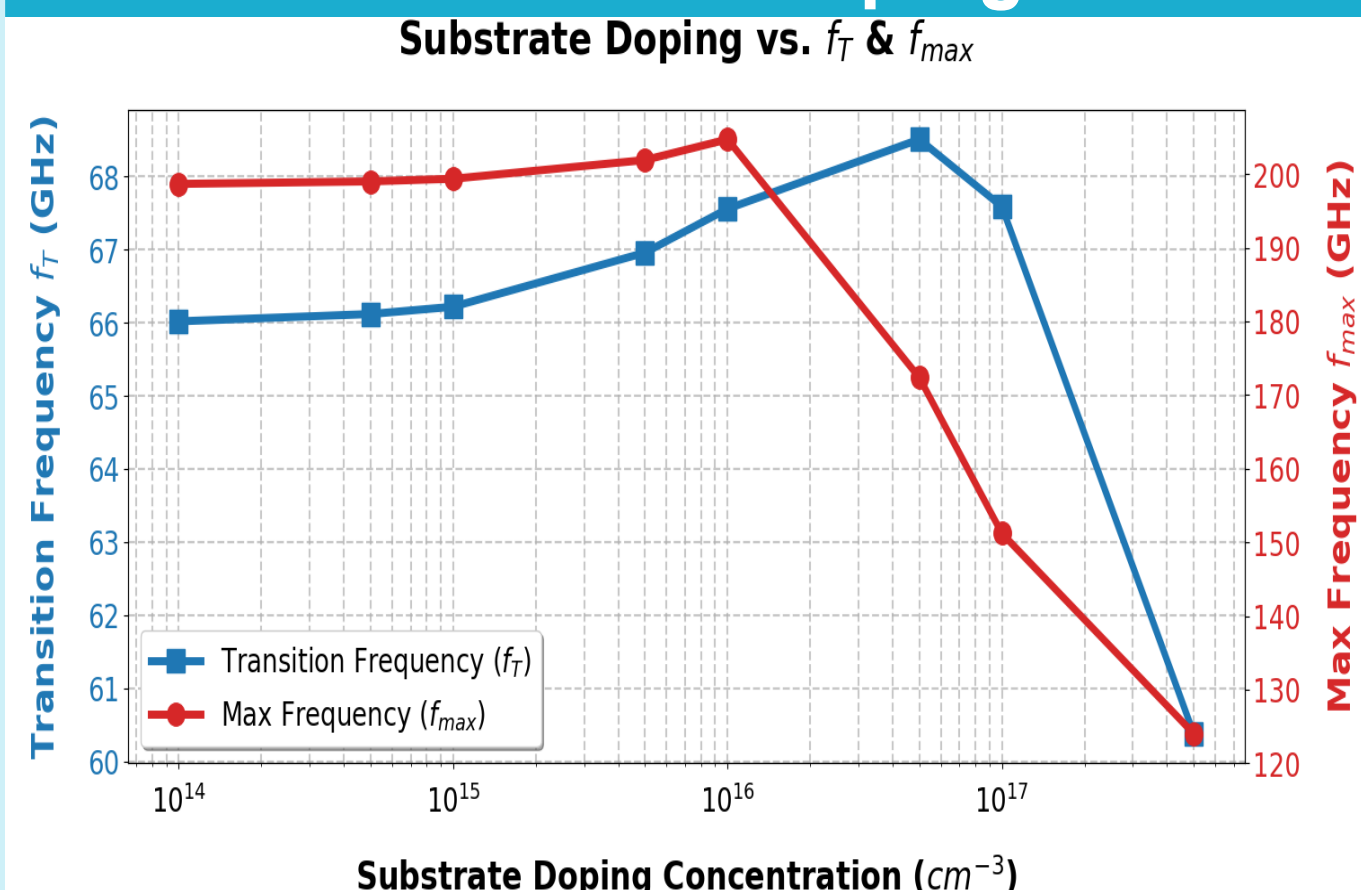


Figure 5: Substrate Doping vs Fmax and Ft.

- Scaling Range:** Changing Substrate Doping from 10^{14} cm^{-3} to $5 \times 10^{17} \text{ cm}^{-3}$.
- Effect:** Initially improves carrier confinement (reducing buffer leakage). At high levels, it increases parasitic substrate capacitance and introduces ionised-impurity scattering, slowing down electrons.
- Observations:** A "bell curve" trend where frequencies peak in the mid- 10^{16} range and then crash as doping exceeds 10^{17} due to substrate conductivity.

Silicon Doping

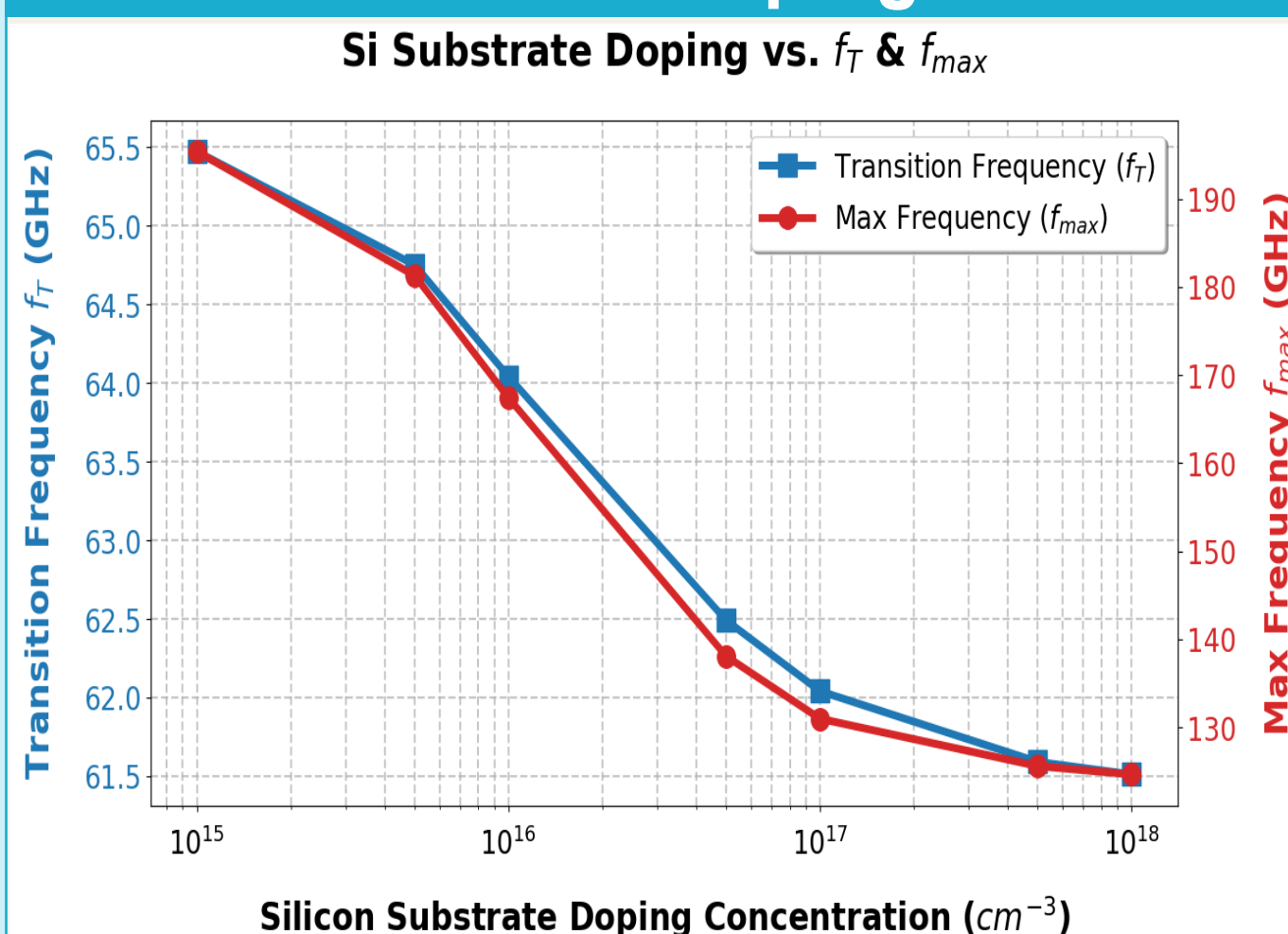


Figure 6: Silicon Doping vs Fmax and Ft

- Scaling Range:** Changing Si doping from 10^{15} cm^{-3} to 10^{18} cm^{-3} .
- Effect:** Worsens performance by increasing substrate loss (RF dissipation) and parasitic capacitive coupling. The substrate acts more like a conductor than an insulator.
- Observations:** A monotonic (steady) decline in both f_T and f_{max} as the substrate becomes more conductive.

Carbon Buffer Doping

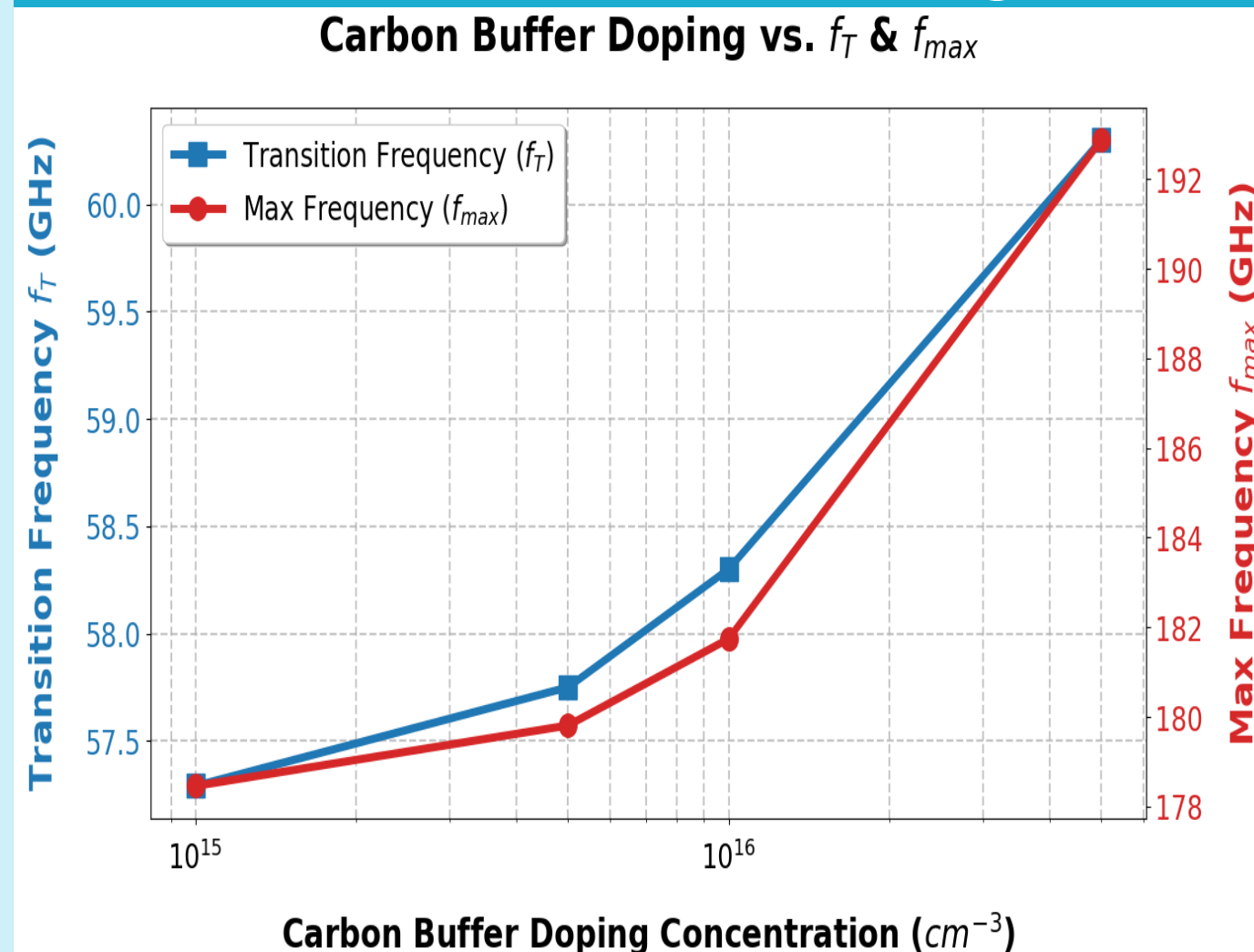


Figure 7: Carbon Buffer vs Fmax and Ft

- Scaling Range:** Changing from 10^{15} cm^{-3} to 10^{17} cm^{-3} .
- Effect:** Improves performance by enhancing channel isolation. It suppresses short-channel effects (SCE) and reduces the output conductance in the high-speed channel.
- Observations:** Both frequencies trend upward as doping increases, showing Carbon is an effective isolation tool for improving speed.

Iron Buffer Doping

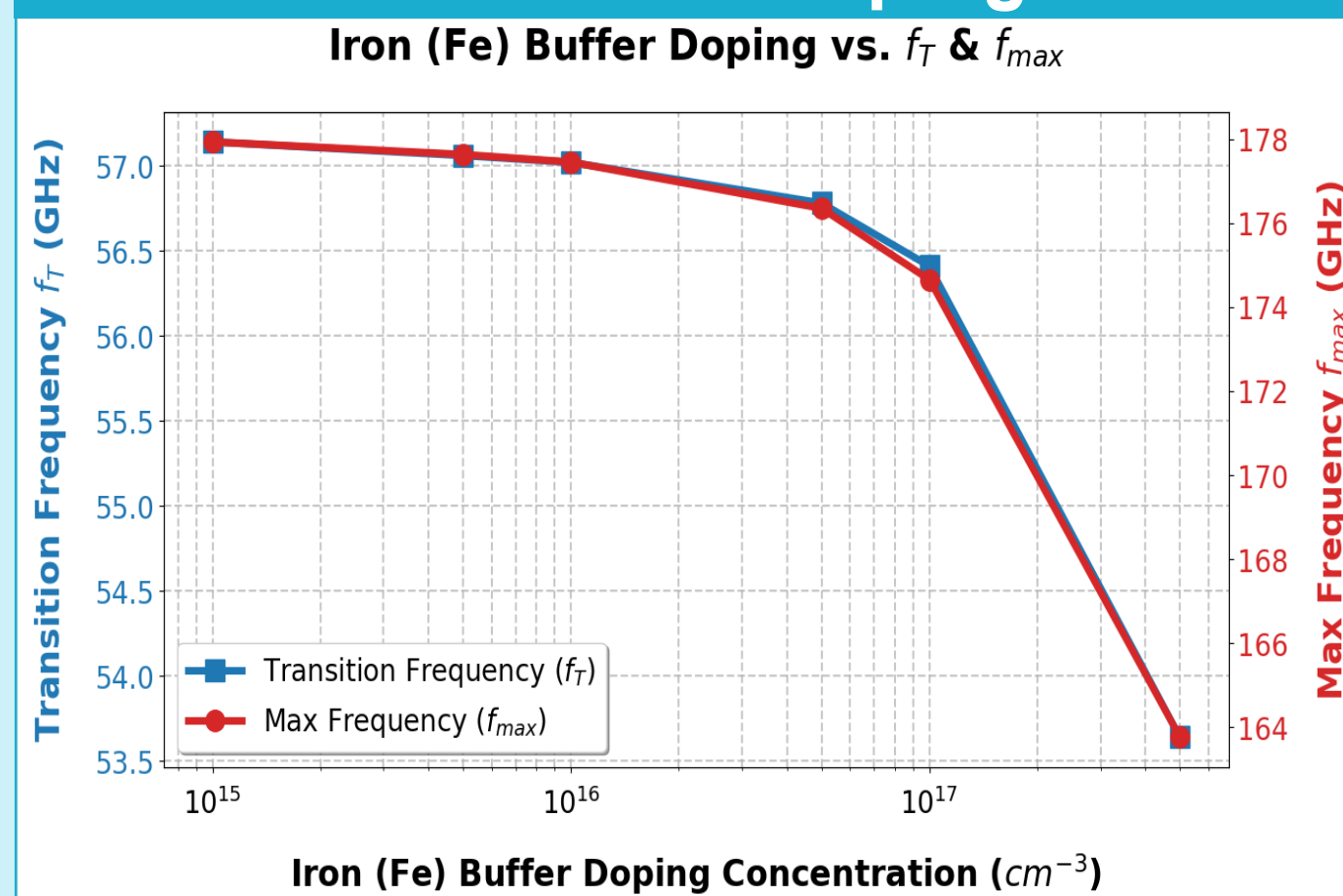


Figure 8: Iron Buffer vs Fmax and Ft

- Scaling Range:** Changing Fe Doping from 10^{15} cm^{-3} to $5 \times 10^{17} \text{ cm}^{-3}$.
- Effect:** Worsens performance at high levels due to deep-level scattering and carrier trapping. This degrades effective electron mobility and introduces signal delays at the channel-buffer interface.
- Observations:** Frequencies remain stable until 10^{16} , then drop sharply as Iron atoms cause carrier scattering and trap-induced delays.

Conclusion

- On decreasing Gate Length, Transition frequency (f_T) and the maximum oscillation frequency (f_{max}) increase.
- Barrier Thickness & $\text{In}_x\text{Ga}_{1-x}$ Mole Fraction:** These elements create a trade-off between Transition Frequency (f_T) and (f_{max}).
- Carbon in buffer** improves both f_T and f_{max} .
- Silicon Doping** in the substrate or **Iron** in Buffer Negatively impacts performance.
- The "Ideal" Configuration:** Optimal performance is achieved by shrinking the gate while balancing doping and trap levels to maximise mobility/isolation and minimize parasitics.

Reference

- Lee J.-H., Ju J.-M., Atmaca G., Kim J.-G., Kang S.-H., Lee Y.-S., Lee S.-H., Lim J.-W., Kwon H.-S., Lişesivdin S.B., et al. High figure-of-merit (V2BR/ON) AlGaIn/GaN power HEMT with periodically C-doped GaN buffer and AlGaIn back barrier. IEEE J. Electron Devices Soc. 2018;6:1179–1186. doi: 10.1109/JEDS.2018.2872975.2.
- Silvestri, M., Uren, M. J., & Kuball, M. (2013). Iron-induced deep-level acceptor center in GaN/AlGaIn high electron mobility transistors: Energy level and cross section. Applied Physics Letters, 102(7), Article 073501. <https://doi.org/10.1063/1.4793196>